PREMKUMAR ADESH MALI

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SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence & Machine Learning, seeking a software engineering or machine-learning internship. Experienced in full-stack web development and real-time computer-vision systems, with hands-on experience in AI-based object detection, REST API development, and database-driven applications. Proven ability to build production-oriented web and computer-vision solutions with measurable improvements in latency and operational efficiency. Eager to contribute technical expertise to a collaborative engineering team while delivering reliable, high-quality software.

EDUCATION

SRM Institute of Science and Technology **CGPA-7.99/10**
B.Tech Aug 2024- May 2028

PROJECTS

SafeVision AI — PPE Detection for Construction Site Safety

- Engineered a React and Tailwind CSS dashboard for live CV inference, achieving sub-100ms latency for safety violation detection.
- Integrated YOLOv8 models via FastAPI to automate real-time site monitoring, successfully flagging 80% of non-compliant PPE cases during testing.
- Streamed detection metrics via Axios, enabling instant visual data rendering for 50+ concurrent site event triggers.
- Directed a two-person engineering team to coordinate AI/frontend workflows, delivering the final solution two weeks ahead of schedule.

BloodLink — Blood Donation Management System

- Architected a normalized SQL schema to manage blood bank inventory, reducing data retrieval time by 30%.
- Developed a full-stack blood donation platform using React and Spring Boot, serving 200+ unique user requests.
- Automated donation lifecycle tracking (Pending to Completed), increasing scheduling efficiency by 40%.
- Engineered a SOS notification system that reduced emergency alert dispatch time by 25% via real-time WebSocket updates.

Unified Banking System — Multi-Bank Aggregator Portal

- Optimized dashboard rendering efficiency using Tailwind CSS, improving page load speed by 20%.
- Developed a secure Spring Boot REST backend to handle transactional data for 500+ user accounts.
- Visualized financial trends using Chart.js, providing users with accurate, real-time insights into spending patterns.
- Secured high-risk operations with JWT authentication and multi-factor re-verification to ensure 70% data integrity.

RationBox — Smart Ration Monitoring App

- Designed a mobile-first Flutter interface that improved user accessibility scores by 15% through Material 3 standards.
- Automated inventory monitoring by building reusable sensor-alert components, reducing manual tracking time by 10 hours weekly.
- Architected a backend data layer that processed multi-sensor inputs with 88% accuracy.

TECHNICAL SKILLS

Tools: VS Code, Git, GitHub
Languages: Python, Java, JavaScript,C, C++, SQL
Frameworks: React.js, Flutter, Spring Boot, Node.js, FastAPI
AI/ML: YOLOv8, OpenCV, NumPy,Pandas
Databases: MySQL, MongoDB, Supabase (Postgres)
Web: Tailwind CSS, Bootstrap, REST APIs, JWT Authentication

INTERESTS

Reading on finance and psychology.